200mW High Speed Switching Diode

Features

- Fast Switching Speed
- High Conductance
- Moisture Sensitivity Level 1
- Epoxy Meets UL 94 V-0 Flammability Rating
- Halogen Free Available Upon Request By Adding Suffix "-HF"
- Lead Free Finish/RoHS Compliant ("P" Suffix Designates RoHS Compliant. See Ordering Information)

Maximum Ratings

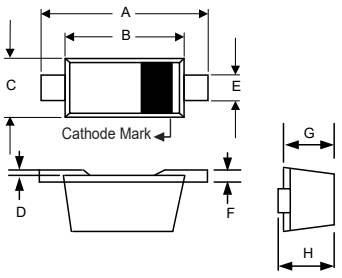
- Operating Junction Temperature Range: -65°C to +150°C
- Storage Temperature Range: -65°C to +150°C
- Thermal Resistance: 833°C/W Junction to Ambient

MCC Part Number	Device Marking
1N4448X	T5

Parameter	Symbol	Value
Non-Repetitive Peak Reverse Voltage	V_{RM}	100V
DC Reverse Voltage	V_R	75V
Peak Repetitive Reverse Voltage	V_{RRM}	75V
Working Inverse Voltage	V_{RWM}	75V
RMS Reverse Voltage	$V_{R(RMS)}$	53V
Forward Continuous Current (Note 1)	I_{FM}	500mA
Average Forward Current	$I_{F(AV)}$	250mA
Peak Forward Surge Current (t=1s) (t=1μs)	I_{FSM}	2A 4A
Power Dissipation	P_D	150mW

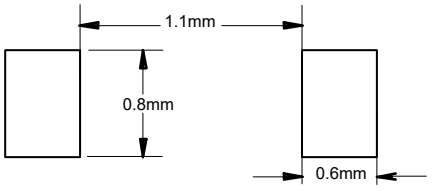
Notes: 1. Valid Provided that Terminals are Kept at Ambient Temperature

SOD-523



DIMENSIONS					
DIM	INCHES		MM		NOTE
	MIN	MAX	MIN	MAX	
A	0.059	0.067	1.50	1.70	
B	0.043	0.051	1.10	1.30	
C	0.030	0.033	0.75	0.85	
D	0.000	0.003	0.00	0.07	
E	0.010	0.014	0.25	0.35	
F	0.003	0.008	0.08	0.20	
G	0.020	0.026	0.50	0.65	
H	0.020	0.026	0.50	0.65	

Suggested Solder Pad Layout



Electrical Characteristics @ 25°C (Unless Otherwise Specified)

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Units
Forward Voltage	V_F	I_F=5mA	0.62		0.72	V
		I_F=10mA			0.855	V
		I_F=100mA			1	V
		I_F=150mA			1.25	V
Reverse Current	I_R	V_R=20V			25	nA
		V_R=75V			2.5	μA
		V_R=25V, T_J=150°C			30	μA
		V_R=75V, T_J=150°C			50	μA
Junction Capacitance	C_J	V_R = 0V, f = 1MHz			4	pF
Reverse Recovery Time	t_{rr}	I_F=I_R=10mA, I_{rr}=0.1×I_R,R_L=100Ω			4	ns

* Pulse Test: Pulse Width 300 μsec, Duty Cycle 2%

Curve Characteristics

Fig. 1 - Typical Instantaneous Forward Characteristics

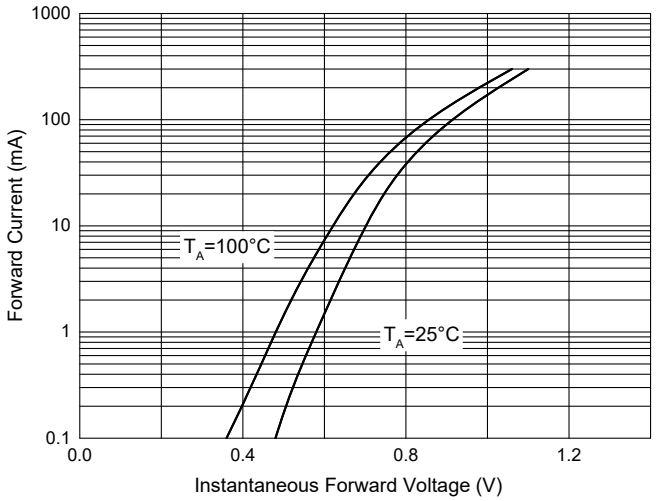


Fig. 2 - Typical Reverse Leakage Characteristics

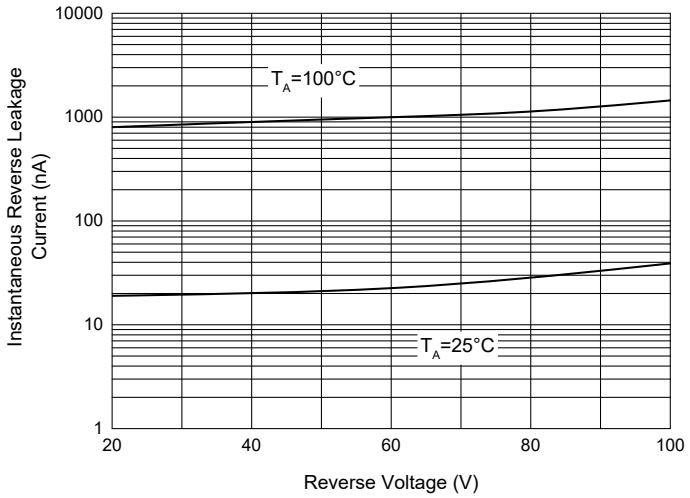
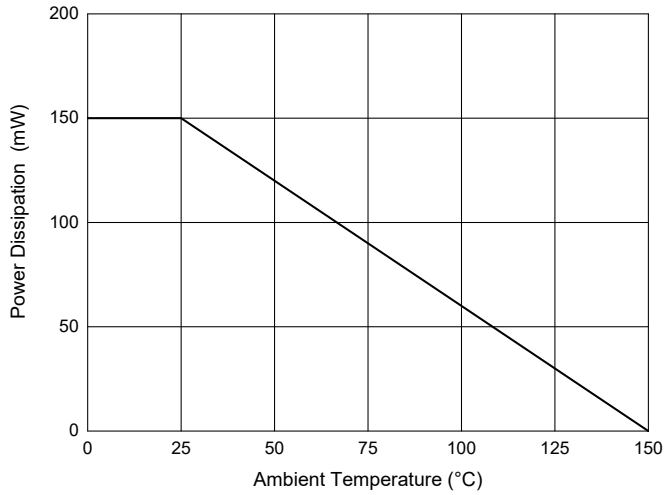


Fig. 3 - Power Derating Curve



Ordering Information

Device	Packing
Part Number-TP	Tape&Reel: 8Kpcs/Reel

Note : Adding "-HF" Suffix For Halogen Free, eg. Part Number-TP-HF

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